

# Applications of Determinants

Lecture 19 and 20

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# Formulas for the Determinant; Applications of Determinants

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# Computation of $A^{-1}$

The 2 by 2 case shows how cofactors go into  $A^{-1}$ :

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{\det A} \begin{bmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \end{bmatrix}.$$

Cofactor matrix  $C$  is transposed

$$A^{-1} = \frac{C^T}{\det A}$$

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} C_{11} & \cdots & C_{n1} \\ \vdots & & \vdots \\ C_{1n} & \cdots & C_{nn} \end{bmatrix} = \begin{bmatrix} \det A & \cdots & 0 \\ \vdots & \vdots & \vdots \\ 0 & \cdots & \det A \end{bmatrix}$$

Note: The above  $C$  should be transposed.

The critical question is: Why do we get zeros off the diagonal?

# Computation of $A^{-1}$

If we combine the entries  $a_{1j}$  from row 1 with cofactors  $C_{2j}$  for row 2, why is the result zero?

$$a_{11}C_{21} + a_{12}C_{22} + \cdots + a_{1n}C_{2n} = 0.$$

- We are computing the determinant of a new matrix  $B$  with a new row 2.
- The first row of  $A$  is copied into the second row of  $B$ . Then  $B$  has two equal rows, and  $\det B = 0$ .

# Example

## Example

The inverse of a sum matrix is a difference matrix:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{has } A^{-1} = \frac{C^T}{\det A} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

The minus signs enter because cofactors always include  $(-1)^{i+j}$ .

# The Solution of $Ax = b$ .

**Cramer's Rule:** The  $j$ th component of  $x = A^{-1}b$  is just  $C^T b$  divided by  $\det A$ . There is a famous way in which to write the answer  $(x_1, x_2, \dots, x_n)$ :

The  $j$ th component of  $x = A^{-1}b$  is the ratio:  $x_j = \frac{\det B_j}{\det A}$ , where

$$B_j = \begin{bmatrix} a_{11} & a_{12} & \cdots & b_1 & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & b_n & \cdots & a_{nn} \end{bmatrix}.$$

Can you prove this rule?

# Example

## Example

The solution of

$$x_1 + 3x_2 = 0$$

$$2x_1 + 4x_2 = 6$$

has 0 and 6 in the first column for  $x_1$  and in the second column for  $x_2$ :

$$x_1 = \frac{\begin{vmatrix} 0 & 3 \\ 6 & 4 \end{vmatrix}}{\begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix}} = \frac{-18}{-2} = 9, \quad x_2 = \frac{\begin{vmatrix} 1 & 0 \\ 2 & 6 \end{vmatrix}}{\begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix}} = \frac{6}{-2} = -3.$$

- The denominators are always  $\det A$ .
- For 1000 equations Cramer's Rule would need 1001 determinants.

# The Volume of a Box

The determinant equals the volume of a box. We first consider right-angled box, which has orthogonal rows:

$$AA^T = \begin{bmatrix} \text{row 1} \\ \vdots \\ \text{row } n \end{bmatrix} \begin{bmatrix} \text{column 1} & \cdots & \text{column } n \end{bmatrix} = \begin{bmatrix} l_1^2 & & 0 \\ & \ddots & \\ 0 & & l_n^2 \end{bmatrix}.$$

The  $l_i$ 's are the lengths of the rows (the edges), and the zeros off the diagonal come because the rows are orthogonal. Using the product and transposing rules,

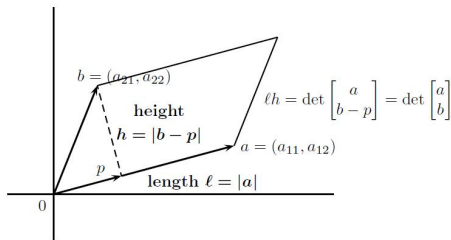
$$l_1^2 l_2^2 \cdots l_n^2 = \det(AA^T) = (\det A)(\det A^T) = (\det A)^2.$$

The square root of this equation says that the absolute value of the determinant of  $A$  equals the volume.



## Not Right-Angle Case

It the angles are not  $90^\circ$ , the volume is **NOT** the product of the lengths. However, by rule 5, like the following figure shows that we can change the parallelogram to rectangle, where it is already that volume = |determinant|.



**Figure 4.2:** Volume (area) of the parallelogram =  $\ell$  times  $h = |\det A|$ .

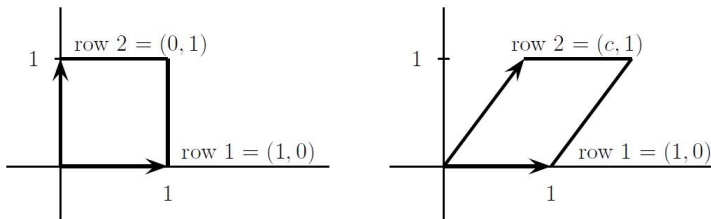
In  $n$  dimensions, the idea is the same. The Gram-Schmidt process produces orthogonal rows, with volume = |determinant|.

# Volume = |Determinant|

We know that

$$\det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1, \det \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix} = 1.$$

These determinants give volumes—or areas, since we are in two dimensions—drawn in Figure 4.3. The parallelogram has unit base and unit height; its area is also 1.



**Figure 4.3:** The areas of a unit square and a unit parallelogram are both 1.

# A Formula for the Pivots

Elimination on  $A$  includes elimination on  $A_2$ :

$$A = \begin{bmatrix} a & b & e \\ c & d & f \\ g & h & i \end{bmatrix} \rightarrow \begin{bmatrix} a & b & e \\ 0 & \frac{ad-bc}{a} & \frac{af-ec}{a} \\ g & h & i \end{bmatrix}$$

Actually it is not just the pivots, but the entire upper-left corners of  $L$ ,  $D$ , and  $U$ , that are determined by the upper-left corner of  $A$ :

$$A = LDU = \begin{bmatrix} 1 & & & \\ \frac{c}{a} & 1 & & \\ * & * & 1 & \end{bmatrix} \begin{bmatrix} a & & & \\ & \frac{ad-bc}{a} & & \\ & & * & \end{bmatrix} \begin{bmatrix} 1 & \frac{b}{a} & * \\ & 1 & * \\ & & 1 \end{bmatrix}$$

# Formulas for pivots

What we see in the first two rows and columns is exactly the factorization of the corner submatrix  $A_2$ . This is a general rule if there are no row exchanges:

## Proposition

*If  $A$  is factored into  $LDU$ , the upper left corners satisfy  $A_k = L_k D_k U_k$ . For every  $k$ , the submatrix  $A_k$  is going through a Gaussian elimination of its own.*

$$LDU = \begin{bmatrix} L_k & 0 \\ B & C \end{bmatrix} \begin{bmatrix} D_k & 0 \\ 0 & E \end{bmatrix} \begin{bmatrix} U_k & F \\ 0 & G \end{bmatrix} = \begin{bmatrix} L_k D_k U_k & L_k D_k F \\ BD_k U_k & BD_k F + CEG \end{bmatrix}.$$

## Formula for pivots

Comparing the last matrix with  $A$ , the corner  $L_k D_k U_k$  coincides with  $A_k$ .

Then:

$$\det A_k = \det L_k \det D_k \det U_k = \det D_k = d_1 d_2 \cdots d_k.$$

The product of the first  $k$  pivots is the determinant of  $A_k$ . This is the same rule we know already for the whole matrix. Since the determinant of  $A_{k-1}$  will be given by  $d_1 d_2 \cdots d_{k-1}$ , we can isolate each pivot  $d_k$  as a ratio of determinants:

$$\frac{\det A_k}{\det A_{k-1}} = \frac{d_1 d_2 \cdots d_k}{d_1 d_2 \cdots d_{k-1}} = d_k$$

# Formulas for pivots

Multiplying together all the individual pivots, we recover

$$d_1 d_2 \cdots d_n = \frac{\det A_1}{\det A_0} \frac{\det A_2}{\det A_1} \cdots \frac{\det A_n}{\det A_{n-1}} = \frac{\det A_n}{\det A_0} = \det A.$$

The pivot entries are all nonzero whenever the numbers  $\det A_k$  are all nonzero:

## Proposition

*Elimination can be completed without row exchanges ( so  $P = I$  and  $A = LU$  ), if and only if the leading submatrices  $A_1, A_2, \dots, A_n$  are all nonsingular.*

# The determinant of a permutation matrix

The determinant of a permutation matrix  $P$  was the only questionable point in the big formula. Independent of the particular row exchanges linking  $P$  to  $I$ , is the number of exchanges always even or always odd?

If so, its determinant is well defined by rule 2 as either 1 or  $-1$ .

- An even number of exchanges can never produce the natural order, beginning with  $(3, 2, 1)$ .
- Let  $N$  count the pairs in which the larger number comes first. Every exchange alters  $N$  by an odd number.

# Determinant of a permutation matrix $P$

- Certainly  $N = 0$  for the natural order  $(1, 2, 3)$ . The order  $(3, 2, 1)$  has  $N = 3$  since all pairs  $(3, 2)$ ,  $(3, 1)$ , and  $(2, 1)$  are wrong.
- We will show that every exchange alters  $N$  by an odd number. Then to arrive at  $N = 0$  (the natural order) takes a number of exchanges having the same evenness and oddness as  $N$ .
- When neighbors are exchanged,  $N$  changes by  $+1$  or  $-1$ .
- Any exchange can be achieved by an odd number of exchanges of neighbors. This will complete the proof; an odd number of odd numbers is odd.



## One Example

- To exchange the first and fourth entries below, which happen to be 2 and 3, we use five exchanges (an odd number ) of neighbors:

$$\begin{aligned} (2, 1, 4, 3) &\rightarrow (1, 2, 4, 3) \rightarrow (1, 4, 2, 3) \rightarrow (1, 4, 3, 2) \\ &\rightarrow (1, 3, 4, 2) \rightarrow (3, 1, 4, 2). \end{aligned}$$

- We need  $l - k$  exchanges of neighbors to move the entry in place  $k$  to place  $l$ .
- Then  $l - k - 1$  exchanges move the originally in place  $l$  (and now found in place  $l - 1$ ) back down to place  $k$ .
- Since  $(l - k) + (l - k - 1)$  is odd, the proof is complete.

# Example

## Example

Find the following determinant:

$$|A| = \begin{vmatrix} 1 & 4 & -1 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 2 & 3 & 11 \\ 3 & 0 & 9 & 2 \end{vmatrix}.$$

# Further Examples

## Example

Suppose the entries of  $A$  satisfy

$$a_{ij} = -a_{ji} \quad (i, j = 1, 2, \dots, n).$$

Find  $\det(A)$  if  $n$  is odd.

**Solution.** 0.

## Example

Find

$$|A| = \begin{vmatrix} a_1 + b_1 & b_1 + c_1 & c_1 + a_1 \\ a_2 + b_2 & b_2 + c_2 & c_2 + a_2 \\ a_3 + b_3 & b_3 + c_3 & c_3 + a_3 \end{vmatrix}.$$

# Example

## Example

Find the determinant:

$$|A| = \begin{vmatrix} a & b & b & \cdots & b \\ b & a & b & \cdots & b \\ b & b & a & \cdots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b & b & b & \cdots & a \end{vmatrix}.$$

**Solution.** Add all the columns to the first column, and factor out an common factor in the first column. Then add  $-1$  times the first row to the rows beneath it to obtain a determinant of an upper triangular matrix, where its determinant can be found by taking the product of the diagonal entries. The determinant of  $A$  is  $[a + (n-1)b](a-b)^{n-1}$ .

# Example

## Example

Find the determinant:

$$|A| = \begin{vmatrix} a & b & & & & \\ c & a & b & & & \\ & c & a & b & & \\ & & \ddots & \ddots & \ddots & \\ & & & c & a & b \\ & & & & c & a \end{vmatrix}.$$

**Solution.**

$$D_n = \begin{cases} \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta}, & \alpha \neq \beta; \\ \frac{(n+1)a^n}{2^n}, & \alpha = \beta. \end{cases}$$

Where  $\alpha + \beta = a, \alpha\beta = bc$ .

## 一些题目

- (1) 设  $A = (a_{ij})$  是三阶非零矩阵,  $|A|$  为  $A$  的行列式,  $A_{ij}$  为  $a_{ij}$  的代数余子式, 若  $a_{ij} + A_{ij} = 0 (i, j = 1, 2, 3)$ , 则  $|A| = \underline{\hspace{2cm}}$ .

(2) 
$$\begin{vmatrix} a & 0 & -1 & 1 \\ 0 & a & 1 & -1 \\ -1 & 1 & a & 0 \\ 1 & -1 & 0 & a \end{vmatrix} = \underline{\hspace{2cm}}$$

- (3) 设  $A = (a_{ij})$  为三阶矩阵,  $A_{ij}$  为元素  $a_{ij}$  的代数余子式, 若  $A$  的每行元素的和均为 2, 且  $|A| = 3$ , 则  $A_{11} + A_{21} + A_{31} = \underline{\hspace{2cm}}$ .

(4)  $n$  阶行列式 
$$\begin{vmatrix} 2 & 0 & \cdots & 0 & 2 \\ -1 & 2 & \cdots & 0 & 2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 2 & 2 \\ 0 & 0 & \cdots & -1 & 2 \end{vmatrix} = \underline{\hspace{2cm}}.$$

# Homework Assignment 19 and 20

4.4: 1, 5, 14, 16, 22, 23, 29, 40.